Process-tensor graphical calculus

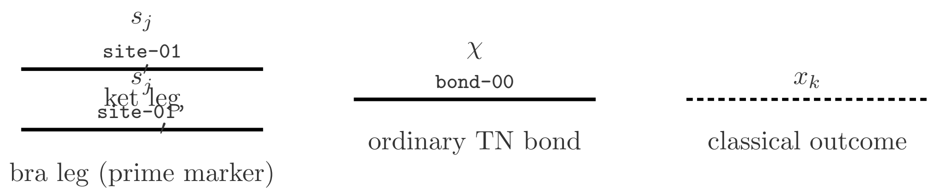
A reusable TikZ library for `ProcessTensors.jl`

Style demo — not a SciPost figure

Time in every process-tensor diagram runs **right to left** ($t_n \leftarrow t_0$). Kets are right-pointing triangles (`----|>`); bras are left-pointing triangles (`<|----`). Spaces are distinguished by line structure, not colour alone.

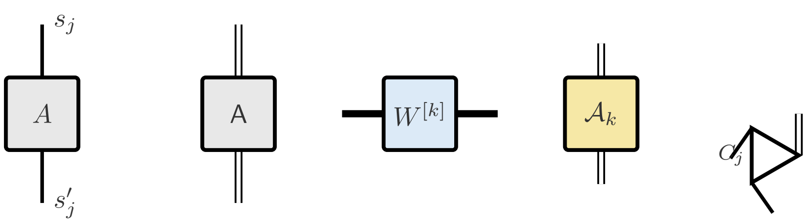
1. Ket and bra triangles

2. Hilbert ket and bra wires

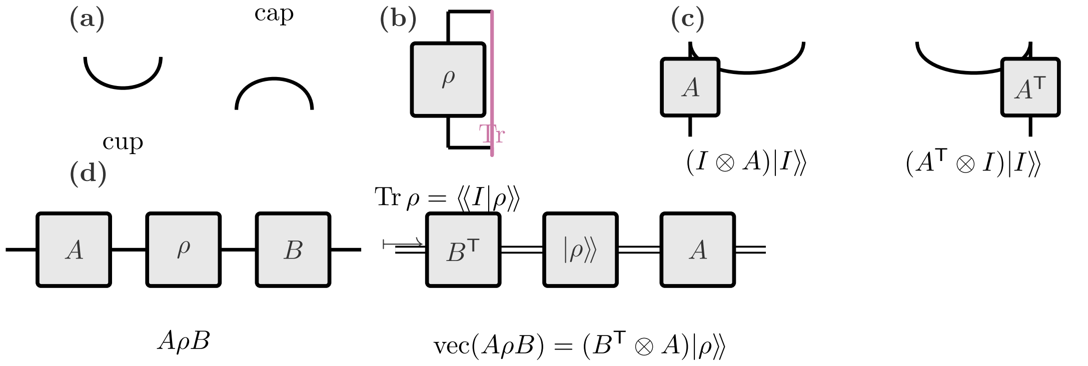


3. Liouville fused wire

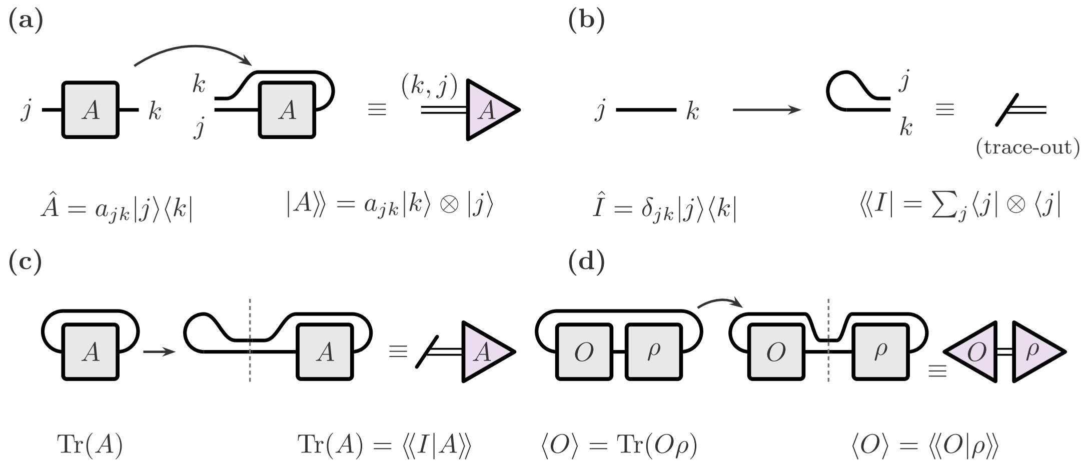
4. Operator and state tensors



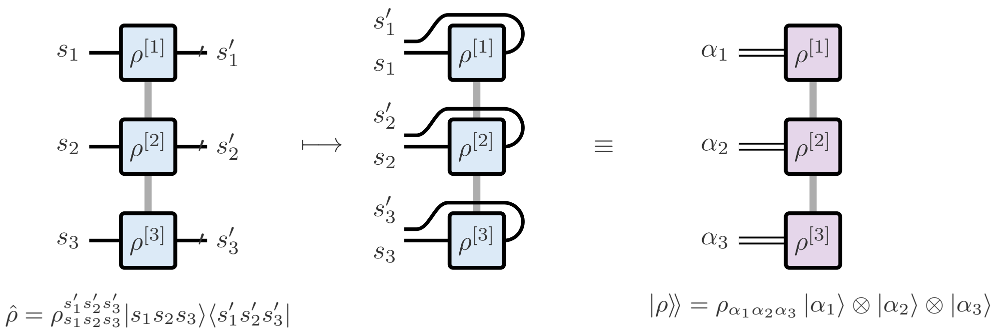
5. Cups, caps, trace closure, and transpose by bending



6. Hilbert operator ↔ Liouville vector

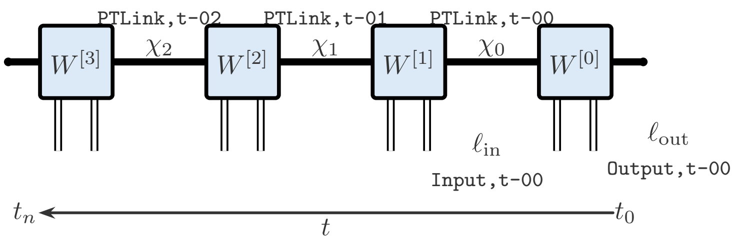


7. Density MPO → Liouville MPS

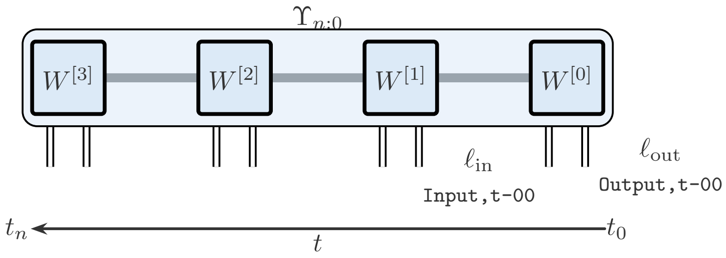


8. PT-MPO with right-to-left time

$$\Upsilon_{n:0} \simeq W^{[n-1]} W^{[n-2]} \cdots W^{[0]}$$

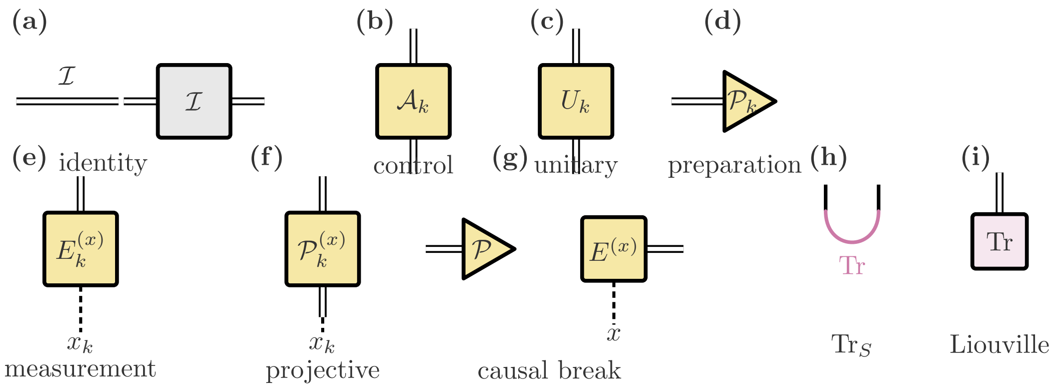


9. Joint non-MPO process tensor

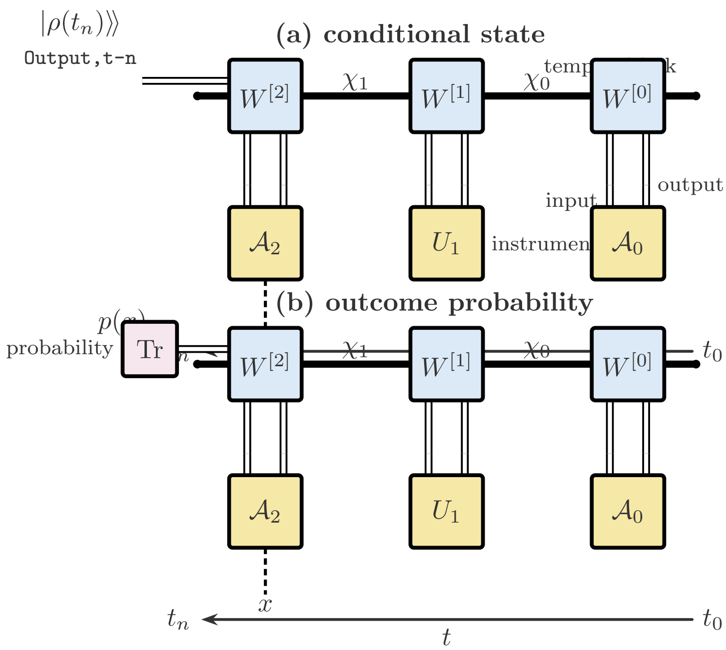


one joint multi-time object (no exposed PT-link indices)

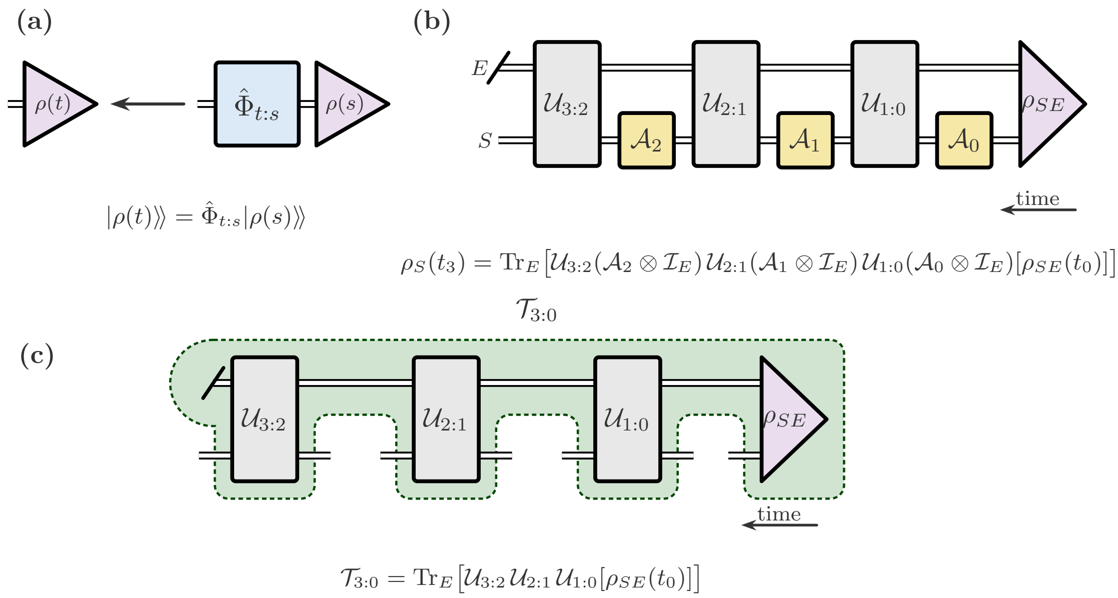
10. Instrument symbols



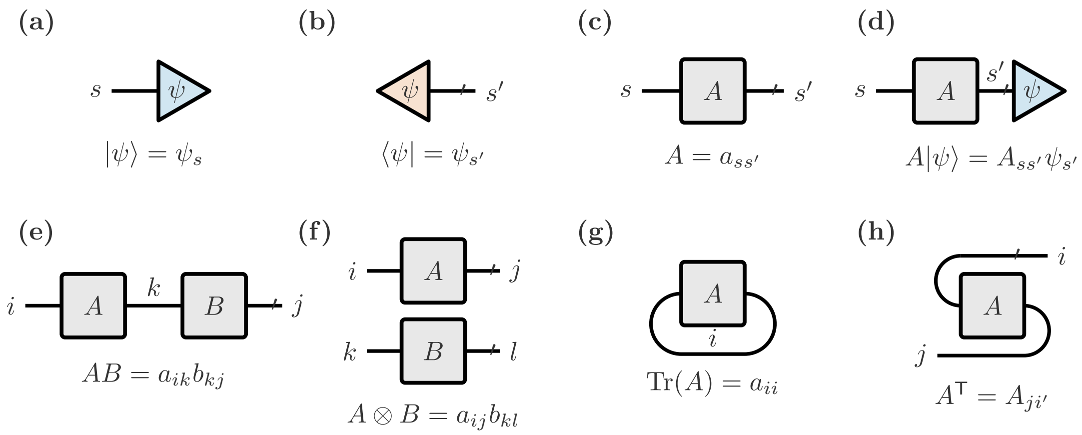
11. PT-MPO contracted with an instrument sequence



12. Channel to multi-time process (Fig. 5, (a)(b) only)



Paper fragment: graphical dictionary



Public commands

Command	Arguments	Purpose
`\PTKet`	name, pos, label	Right-pointing ket triangle.
`\PTBra`	name, pos, label	Left-pointing bra triangle.
`\PTLiouvilleKet`	name, pos, label	Liouville ket (double-wired leg).
`\PTLiouvilleBra`	name, pos, label	Liouville bra (double-wired leg).
`\PTLiouvilleKetLeft`	name, pos, label	Left-pointing Liouville ket (RTL output).
`\PTTensor`	[keys] name pos label	Generic square core, thick rounded stroke.
`\PTOperator`	[keys] name pos label	Hilbert-space operator (square).
`\PTSuperoperator`	[keys] name pos label	Superoperator core (square).
`\PTProcessCore`	[keys] name pos label	PT-MPO time core.
`\PTInstrument`	[keys] name pos label	Instrument core (gold fill).
`\PTInstrumentNode`	[keys] name pos label	Generic instrument constructor.
`\PTIdentity`	[keys] name pos label	Identity box $\mathcal{I}$.
`\PTControl`	[keys] name pos label	General control $\mathcal{A}_k$.
`\PTUnitary`	[keys] name pos label	Unitary U_k.
`\PTPrepare`	[keys] name pos label	Preparation (triangle).
`\PTMeasure`	[keys] name pos label	Measurement with classical x.
`\PTProjective`	[keys] name pos label	Projective instrument.
`\PTCausalBreak`	[keys] name pos	Measure then prepare.
`\PTTraceHilbert`	ket-coord, bra-coord	Magenta Hilbert-space trace cup.
`\PTTraceLiouville`	name, pos, label	Compact Liouville Tr node.
`\PTCombiner`	[dir] name pos label	Fuse $s \otimes s' \to \ell$.
`\PTCombinerInv`	[keys] name pos label	Inverse combiner.
`\PTIndexLabel`	[keys] coord	Math + ITensor tag + prime.
`\PTCup` / `\PTCap`	[style] A B	Smooth cup / cap.
`\PTTimeArrow`	[anchor] right left label	Left-pointing time arrow.
`\PTDrawProcessMPO`	[keys]	Arbitrary-n PT-MPO.
`\PTDrawJointProcess`	[keys]	Joint (non-MPO) process.
`\PTOperatorLegs`	node	Ket (north) and bra (south) stubs.
`\PTCallout`	[keys] coord text	Sparse annotation.

`\PTInstrumentNode` keys: `kind=box|ket|bra|trace`, `fill=`, `classical=none|up|down|left|right`, `classical label=`, `minimum width=`, `minimum height=`.

`\PTDrawProcessMPO` keys: `n=`, `at=(x,y)`, `name=`, `spacing=`, `joint=true|false`, `tags=`, `chi=`, `times=`, `time arrow=`, `final=`, `label=`.

`\PTIndexLabel` keys: `anchor=`, `math=`, `tag=`, `prime=`, `xshift=`, `yshift=`.